

Project Title: EdTech Student Performance & Engagement Portal

1. Project Introduction:

The EdTech Student Performance & Engagement Portal is a data analytics system designed to monitor, analyze, and visualize student academic performance and engagement in online learning environments. The portal collects data from various educational activities such as course completion, assignment submissions, quizzes, assessments, attendance records, and login activities.

The main purpose of the project is to help educational institutions identify student learning patterns, improve academic outcomes, and provide early intervention for students who may be struggling academically.

2. Problem Statement:

Educational institutions generate large amounts of student data every day. However, this data often remains unused due to the lack of proper analysis tools.

Some common challenges include:

- Difficulty tracking student engagement.
- Inability to identify students at risk of poor performance.
- Lack of centralized monitoring systems.
- Limited insights into learning behaviors.
- Manual analysis consuming significant time and effort.

The proposed portal addresses these challenges by providing a centralized dashboard that visualizes student performance and engagement metrics.

3. Project Objectives:

The primary objectives of this project are:

Academic Performance Tracking:

- Monitor student grades.
- Analyze quiz and examination results.
- Compare performance across courses.

Student Engagement Monitoring:

- Track login frequency.
- Measure course completion rates.
- Monitor assignment submission patterns.

Data Visualization:

- Generate graphs and charts.
- Display performance trends.
- Create interactive dashboards.

Early Warning System:

- Identify low-performing students.
- Generate alerts for reduced engagement.
- Support faculty decision-making.

4. Scope of the Project:

The portal can be used by:

Students:

- View academic progress.
- Monitor learning activities.
- Track assignment status.

Faculty Members:

- Analyze class performance.
- Monitor student participation.
- Identify students needing support.

Administrators:

- Generate institutional reports.
- Monitor department performance.
- Make data-driven decisions.

5. Data Sources:

The system can use data collected from:

Student Information:

- Student ID
- Name
- Branch
- Semester

Attendance Data:

- Attendance percentage
- Number of absences

Assignment Data:

- Submission date
- Completion status
- Assignment scores

Assessment Data:

- Quiz scores
- Mid-term marks
- Final examination marks

Engagement Data:

- Login frequency
- Time spent on portal
- Number of completed modules

6. Key Features:**1. Student Dashboard:**

Displays:

- Total marks
- Attendance percentage
- Course progress
- Assignment completion rate

2. Faculty Dashboard:

Displays:

- Class performance
- Average marks
- Student engagement metrics

3. Performance Analytics:

Provides:

- Grade distribution
- Subject-wise analysis
- Semester-wise comparison

4. Engagement Analytics:

Provides:

- Daily logins
- Weekly activity reports
- Course completion statistics

5. Alert System:

Automatically identifies:

- Low attendance students
- Missing assignments
- Poor academic performance

7. Technologies Used:

Development Environment: Google Colab

Libraries:

Pandas:

Used for:

- Data cleaning
- Data manipulation
- Data analysis

NumPy:

Used for:

- Numerical computations
- Statistical calculations

Matplotlib:

Used for:

- Graph generation
- Performance visualization

Seaborn (Optional):

Used for:

- Advanced visualizations

8. Project Workflow:

Step 1: Data Collection:

Collect student academic and engagement data.

Step 2: Data Cleaning:

Remove:

- Missing values
- Duplicate records
- Incorrect entries

Step 3: Data Analysis:

Analyze:

- Attendance trends
- Assignment completion rates
- Student performance patterns

Step 4: Data Visualization:

Create:

- Bar charts
- Pie charts
- Line graphs
- Histograms

Step 5: Dashboard Development

Present findings through interactive dashboards.

Step 6: Reporting:

Generate reports for faculty and administrators.

9. Sample Insights Generated

The portal can answer questions such as:

- Which students have low attendance?
- Which students submit assignments late?
- Is there a relationship between attendance and grades?
- Which course has the highest completion rate?
- Which students are at risk academically?

10. Expected Outcomes:

After successful implementation, the system will:

- Improve student monitoring.
- Increase academic performance.
- Enhance student engagement.
- Reduce dropout risks.
- Support data-driven educational decisions.